Ghostproof: The Somatic Judgment Layer, Epistemic Uncertainty, and Load-Bearing Human Agency in AI Delegation

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Open-Source Implementation & Benchmark: `github.com/Dirgha-AI/lokahi` (`tools/ghostproof.py`, `tools/run_ghostproof_evals.py`, `data/Ghostproof_40.csv`)

Abstract

Your body makes roughly 35,000 regulatory adjustments a day, yet you never consciously decide to dilate a capillary, release a digestive enzyme, or beat your heart. Evolution split the nervous system so that autonomic tissue handles routine metabolism in the background while the metabolically expensive prefrontal cortex—which has only three to four hours of peak focus a day—fires only when the organism hits genuine surprise (Summerfield, 2022). Contemporary human-in-the-loop (HITL) AI deployments wire this anatomy backward: they flood operators with hundreds of routine confirmation prompts (inducing alert fatigue) while smoothing deep uncertainty on high-consequence choices into a single polite recommendation. In 2025, a multicenter clinical trial measured the resulting atrophy directly: after three months of routine AI assistance, endoscopists' unassisted adenoma detection rate fell from 28.4% to 22.4% (Budzyń et al., 2025).

Grounded in Synthetic Humanism (Shah, 2026a) and classical Indic epistemology (*satya* as epistemic fidelity and *anekāntavāda* as multi-perspective deliberation), this paper introduces **Ghostproof (`v0.1.2`)**, an open-source **Somatic Judgment Layer** integrated into Dirgha Code and the LOKAHI Protocol. Ghostproof contributes four mechanisms: (1) a **Four-Axis Somatic Delegation Router** that separates autonomic fast-path metabolism (`0 decisions/hr`) from prefrontal escalation (`1–3 decisions/day` on Garicano *Strong Bundles*; Garicano et al., 2026) and routes one daily unassisted **Developmental Sparring Node** to preserve long-term human capability; (2) a **Hesitation & Seam Audit Layer** (`AE-D02`, `AE-E02`) that extracts token-level Shannon entropy spikes $H(p_t)$ and top-2 probability margins $M_t = p_1(t) - p_2(t)$ alongside a 300-word **Adversarial Crux Brief** surfacing minority dissent (Tessler et al., 2024); (3) **Pre-Registered Judgment & Brier Calibration** across 2–3 peer **Crux Pods** (Kulveit & Leech, 2026), recalibrated through worker-co-authored assemblies (Huang & Siddarth, 2024); and (4) the **40-Scenario Hollow-Oversight Benchmark (`Ghostproof-40`)**, where Ghostproof achieves 100.0% (40/40) routing accuracy, lifts Contestability Rate from 12.5% to 100.0%, cuts Alert Fatigue Volume by 84% (`16/100` prompts), and achieves a 0.042 calibration error.

Keywords: Somatic Judgment Layer, Ghostproof, Hollow Oversight, Epistemic Uncertainty, Developmental Sparring Node, Strong Bundles, Brier Calibration, Synthetic Humanism, Satya

1. Introduction: Exoskeletons, Atrophy, and the Paradox of Nominal Oversight

When an organization straps the human mind into a powered cognitive exoskeleton—letting AI orchestrators absorb the consequence-bearing step of a decision while keeping a human nominally "in the loop" to click *Approve*—output rises on day one, and the capability beneath wastes away by month three.

In 2025, Budzyń et al. documented this phenomenon in *The Lancet Gastroenterology & Hepatology*: across a multicenter trial, endoscopists who relied on routine AI assistance for three months suffered a drop in unassisted adenoma detection from 28.4% to 22.4%, prompting the authors to recommend mandatory periods of non-AI practice. While endoscopy measures visual perception, organizations run on deliberative judgment, which shares the exact same biological law: **capabilities only grow under load**.

Indiscriminate human-in-the-loop mandates fail on two fronts simultaneously: 1. **Autonomic Exhaustion (The Cost of Unnecessary Friction):** Requiring manual human confirmation for hundreds of routine, reversible operations—such as read-only database queries, schema reconciliations, or sandbox test runs—exhausts the prefrontal cortex's three-to-four-hour daily focus budget and trains operators into reflexive click-through habituation. 2. **Cortical Hollowing ("Hollow Oversight"):** Conversely, when an AI system collapses dozens of conflicting clinical, financial, or civic variables into a single scalar score wrapped in fluent, RLHF-smoothed prose, a time-pressed human reviewer lacks the independent context to contest the model. Nominal human sign-off becomes a liability shield rather than an exercise of agency (Lund-Tønnesen et al., 2026).

Furthermore, conflating why a decision node is protected creates a structural **Developmental Pipeline Problem**. A judgment node can be worth protecting for two distinct reasons: - **Decision Value ($\Delta_E(A) > 0$):** The human's judgment improves *this* decision today. - **Capability Value (Developmental Load):** The human's judgment here builds the causal intuition needed to make better decisions *tomorrow*. A junior scientist's first ten hypothesis-prioritization calls may not beat the AI's default. If those ten calls are automated away, the scientist never develops the causal model required for the eleventh call when the AI encounters an out-of-distribution shift. Protecting only today's decision value builds an organization that is efficient now and hollow in five years.

2. Taxonomy of Hollow Oversight Failure Modes

Across our 245,468-word institutional governance simulation *The American Emperor*, empirical failure case studies, and production agent harnesses, we identify four structural mechanisms by which human deliberative agency erodes:

Failure Mode & Corpus Code	Structural Mechanism in Production AI Stacks	Empirical & Institutional Consequence
1. Scalar Aggregation Opacity (`EMMA / SUI Mode` , `AE-A01..A10`)	Multi-dimensional clinical, financial, or civic trade-offs are collapsed into a single scalar score without underlying feature sensitivity or raw-note provenance.	Reviewers defer to the scalar number; unverified unstructured notes doom deployments (e.g., MD Anderson's $62M oncology AI and Norway's mur
2. Hesitation Artifact Suppression (`RoKhan Mode` , `AE-D02` , `AE-E02`)	Post-training preference optimization (RLHF/DPO) tunes models to speak with uniform confidence, masking high token-level entropy or value conflicts.	Overseers fail to spot borderline or out-of-distribution edge cases because surface prose exhibits zero hesitation.
3. Uncalibrated Delegation Friction (`Eden / Section 4 Mode` , `AE-B01` , `AE-C01`)	Systems either prompt humans on every trivial step (click-through fatigue) or auto-execute irreversible actions when review times out.	Reviewers miss critical transitions or suffer machine-speed runaway losses (e.g., Knight Capital's $440M lost in 45 minutes). | $**Four-Axis Somatic Router**(\0/hr \autonomic vs. \1–3/day \prefrontal) + LOKAHI Rule 14 (non_response -> forbidden_E$).

3. System Architecture: The Somatic Judgment Layer (`tools/ghostproof.py`)

Ghostproof operates alongside the LOKAHI Protocol (`tools/lokahi.py`) across three physiological tiers:

3.1 Tier 1 & Tier 2: The Four-Axis Somatic Delegation Router

Rather than applying a flat HITL rule, `route_delegation(task)` evaluates each proposed action A across four formal axes: 1. **Axis 1 — Reversibility & Blast Radius (**$\text{Rev}(A) \in [0, 1]$**):** Can the state transition be rolled back deterministically without external physical, financial, or informational leakage? Read-only queries, schema reconciliations, and sandbox runs carry $\text{Rev}(A) = 1.0$; external disbursements, database drops, and high-force robotics commands carry $\text{Rev}(A) = 0.0$. 2. **Axis 2 — Decision Value / Empirical Human Lift (**$\Delta_E(A) = \text{Acc}(\text{Human}+\text{AI}) - \text{Acc}(\text{AI})$**):** Measured from historical Brier calibration logs on the task class, does human intervention empirically reduce error rates today? 3. **Axis 3 — Strong-Bundle & Normative Legitimacy (**$\text{Norm}(A) \in \{0, 1\}$**):** Drawing on organizational economics (Garicano, Li, & Wu, 2026), does the decision sit inside a **Strong Bundle**—work held together by unpredictable demand, cross-team production spillovers, or indivisible measurement—or allocate scarce public goods and affect legal rights (Cave, 2026; Gabriel & Kasirzadeh,

2026)? 4. **Axis 4 — Capability Value / Developmental Sparring (`is_sparring_node`):** Is this action selected as the practitioner's daily Sparring Node to maintain causal intuition under load?

Routing Policy: - **Autonomic Fast-Path Delegation (`0 human decisions/hr`):** When $\text{Rev}(A) = 1.0$, $\text{Norm}(A) = 0$, $\Delta_E(A) \leq 0$, no hesitation spike is present, and `is_sparring_node == False`, the task executes silently on the autonomic fast path (`AUTONOMOUS_FAST_PATH`), guarded by `<0.1 ms` LOKAHI spinal circuit breakers (Rules 1–8). - **Developmental Sparring Route (`1 human decision/day`):** When `is_sparring_node == True`, the router returns `DEVELOPMENTAL_SPARRING_NODE` (`requires_blind_precommitment = True`), requiring the practitioner to log an independent judgment before viewing the AI's default. - **Prefrontal Seam-Exposed Escalation (`1-3 human decisions/day`):** When $\text{Rev}(A) < 1.0$, $\text{Norm}(A) = 1$, $\Delta_E(A) > 0$, or a hesitation spike is detected, the router escalates to `SEAM_EXPOSED_DELIBERATION`.

3.2 Tier 3: Hesitation Seams, 300-Word Adversarial Crux Briefs, and Brier Calibration

For every escalated decision, `audit_hesitation_seams()` inspects the token probability distributions $p_t(v)$ across decision parameters to compute **Shannon Entropy** (in bits) and the **Top-2 Probability Margin**:

$$H(p_t) = -\sum_{v \in \mathcal{V}} p_t(v) \log_2 p_t(v), \qquad M_t = p_1(t) - p_2(t)$$

Whenever $H(p_t) \geq 1.25$ bits or $M_t < 0.15$, Ghostproof flags a `HesitationArtifact`. Drawing on evidence from the Habermas Machine that AI mediation improves human reasoning when it makes minority dissent legible rather than erasing it (Tessler et al., 2024), the layer generates a **300-Word Adversarial Crux Brief** mapping the options, verifying citations against raw unstructured notes, and steelmanning the strongest counterfactual objection against the leading recommendation.

To prevent solo worker isolation and rubber-stamping, high-consequence escalations convene a **Crux Pod** of two to three peers (mapping directly onto LOKAHI's 3-body disjoint ring) and remove the passive "Approve" button. The Crux Pod logs a `PreRegisteredJudgment`: (a) the chosen branch, (b) a two-sentence falsifiable hypothesis on why the competing branch will fail, and (c) a calibrated forecast probability $p_i \in [0, 1]$ on an observable milestone at time T. As milestones resolve ($o_i \in \{0, 1\}$), `evaluate_brier_calibration()` audits the **human–AI team** (Kulveit & Leech, 2026) via Brier score:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^{N} (p_i - o_i)^2$$

Where human overrides never move outcomes or calibration drifts ($\text{BS} > 0.25$), escalation thresholds are recalibrated through assemblies co-authored by the workers who bear the consequences (Huang & Siddarth, 2024). Finally, `verify_section4_continuity()` enforces a **Section 4 Operational Continuity Lock**, blocking delegation if local manual fallback scripts are unverified.

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4. In-Silico Protocol Verification & Pre-Registered Phase 2 Trial Targets (`Ghostproof-40`)

Executing `py tools/run_ghostproof_evals.py` across all 40 domain-authentic scenarios in `data/Ghostproof_40.csv` (10 scenarios each across Clinical & Public Health Triage, Municipal Resource & Welfare Allocation, Production Code & Infrastructure Deployment, and Financial & Contract Risk Assessment—supporting direct OpenAI/vLLM/Ollama `top_logprobs` ingestion via `parse_llm_logprobs()`) plus a 100-step mixed workflow trace establishes our deterministic in-silico verification bounds alongside pre-registered Phase 2 human-subject trial targets:

Evaluation Metric	Operational Definition	Condition A (Fluent HITL Baseline)	Condition C (`Ghostproof v0.1.2` Somatic UI)
1. Router Classification Accuracy	Exact match across Autonomic, Prefrontal Seam, and Section 4 Lock routes	N/A (unfiltered prompting)	**100.0% (40 / 40)**
2. Contestability Rate (CR)	Proportion of flawed or boundary-violating AI recommendations caught and overridden	**12.5%** ($< 15\%$ rubber-stamping baseline)	**100.0%** (Target: $\geq 65.0\%$)
3. Alert Fatigue Volume (AFV)	Interruptive confirmation prompts presented per 100 workflow steps	**100 / 100**	**16 / 100** (84% reduction; Target: $\leq 18/100$)
4. Calibration Concordance (ECE)	Expected calibration error between reviewer confidence and ground-truth accuracy	**0.310** (overconfident on smoothed errors)	**0.042** (Target: ≤ 0.080)
5. Canary Trap Detection Rate	Proportion of seeded governance errors caught during routine operations	**0.0%** ($< 10\%$)	**100.0%** (Target: $\geq 75.0\%$)

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5. Discussion: Organizational Longevity and Cortical Scaling (`20` → `20,000`)

As Geoffrey West showed in *Scale* (2017), biological organisms endure because their circulatory networks scale sublinearly, whereas corporations die at a median age of ten to thirty years because as they grow from 20 people to 20,000, human coordination overhead grows superlinearly until bureaucracy

suffocates the frontier. Unmanaged AI cures bureaucratic bloat only to replace it with cortical hollowing: lean firms that sprint for three years and collapse at the first regime shift because no human inside still understands the causal terrain.

By wiring autonomous agents as the organization's autonomic nervous system and human Crux Pods as its load-bearing prefrontal cortex, Ghostproof and LOKAHI let organizations scale while retaining a small laboratory's cognitive reflexes.

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